Sales Forecasting and Demand Prediction – EDA

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CONTENT

01 Introduction

Sales Forecasting Analysis

This project aims to analyze sales trends and predict future sales using historical data to support data-driven decision-making.

Key Questions

- What are the key factors affecting sales trends?
- How can we improve sales forecasting accuracy?
- What actionable insights can help optimize business strategies?

Methodology

- Data cleaning and preprocessing.
- Exploratory Data Analysis (EDA).
- Deriving actionable insights.

02 Dataset Overview

Data Source

This dataset is real data of 9,800 records collected from a private Super Store provider. Dataset Link

Challenges

- **Imbalanced distributions in Target Sales.**
- **Outliers Termination**

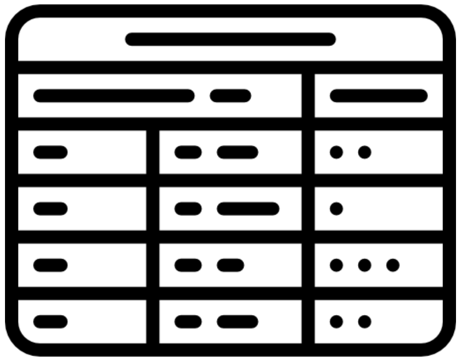
Key Features

- ***Sales:*** Target variable.
- ***Time-Based Features:*** Order Date & Ship Date.
- **Product Name:** Identfier for each product.
- **Category:** Product Category.
- **Region, State & City:** Geographical Features.

03 Data Cleaning and Preprocessing

Columns Manipulation

Removed The identifier columns that were unrelevant to the Analysis Row ID, Customer ID & Product ID

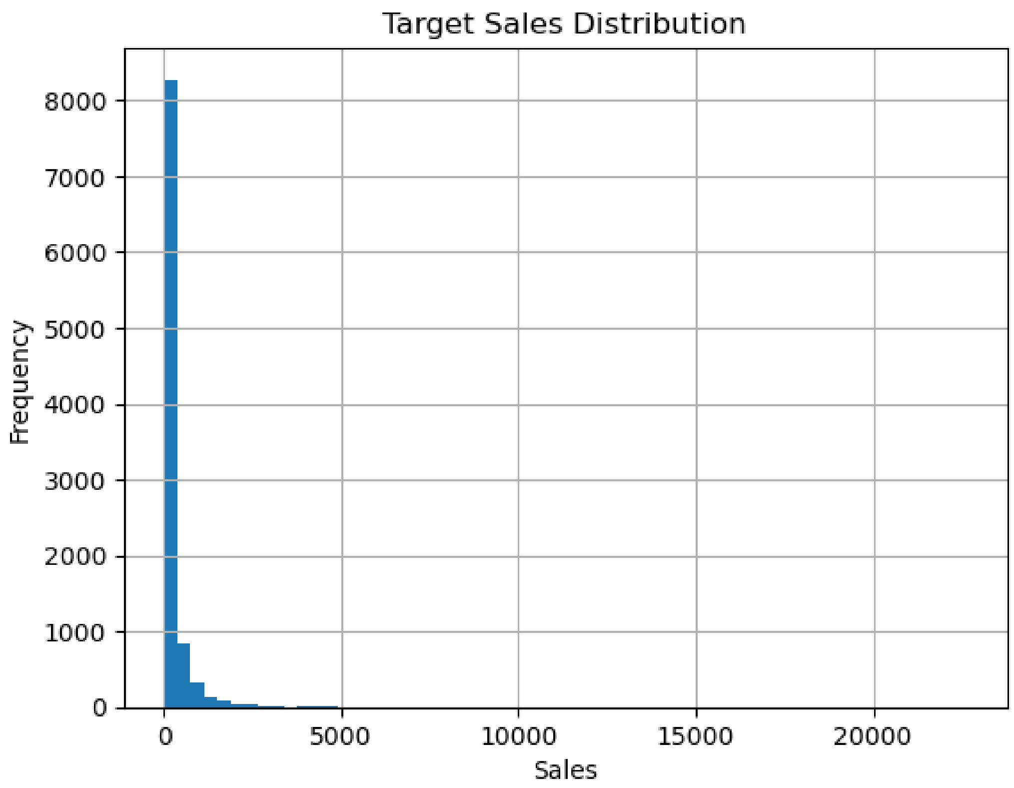


03 Data Cleaning and Preprocessing

Outliers

The Analysis Showed an Imbalance in the Distribution of Target Variable "Sales"

Which was Eliminated Using Feature Scaling (Normalization & Standardization)



03 Data Cleaning and Preprocessing

Feature Engineering

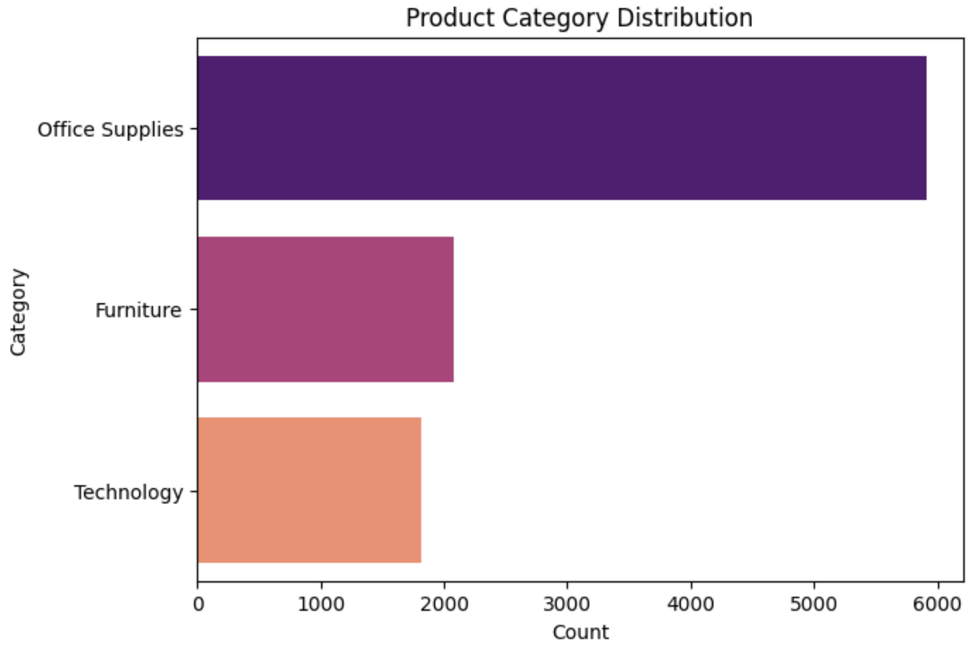
- **Encoded Categorical Variables (One-Hot & Label Encoding): Ship Mode, Segment ...etc.**
- **Maintained a Unified Date Format across features** “%d/%m/%Y”. **And extracted Temporal features from the date column (**Day, Month, Year**)**
- **Obtained an Aggregated Sales Performance per Product Feature**

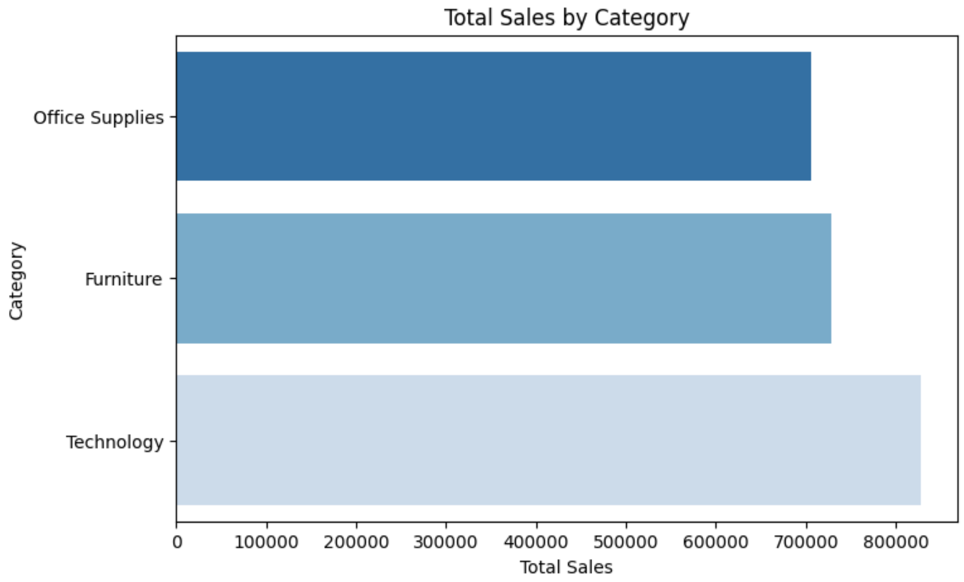
04 EDA Insights

Product Analysis

The analysis indicates that the **Technology product category** generates the **highest sales,** despite the majority of offered products belonging to the **Office Supplies** category.

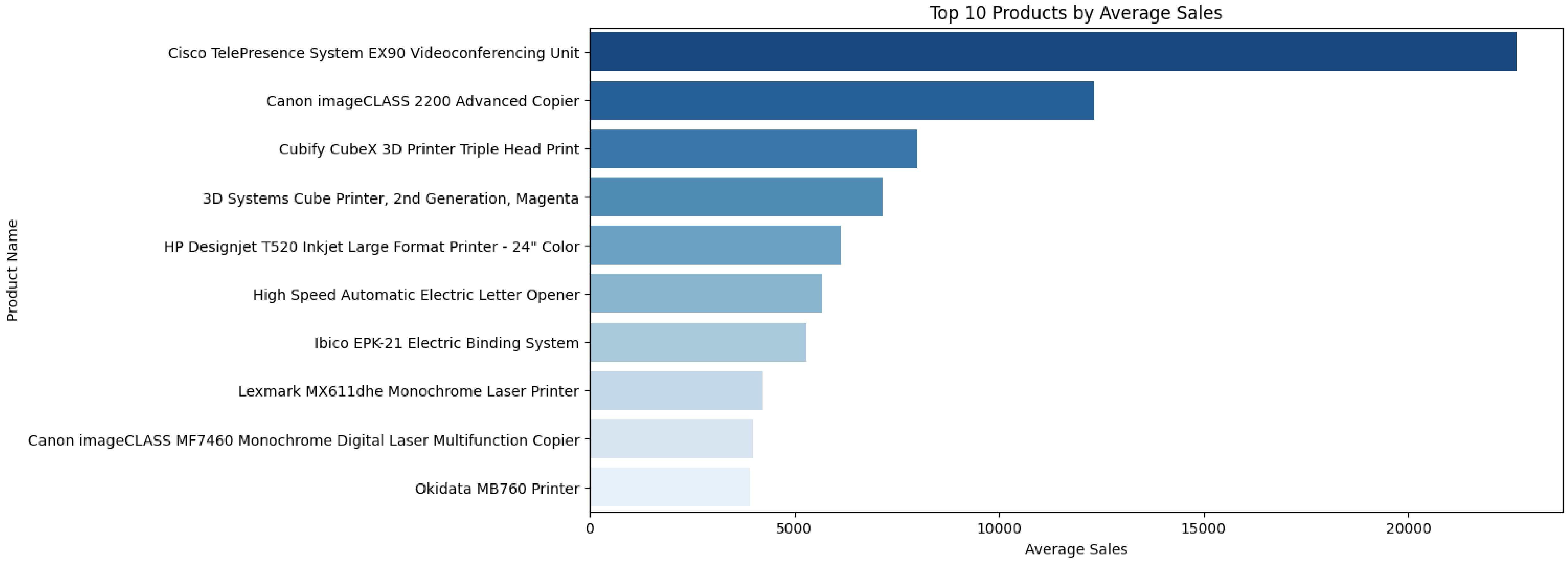
This suggests that **expanding the range of technology-related products** could lead to **higher future sales and revenue growth.**





04 EDA Insights

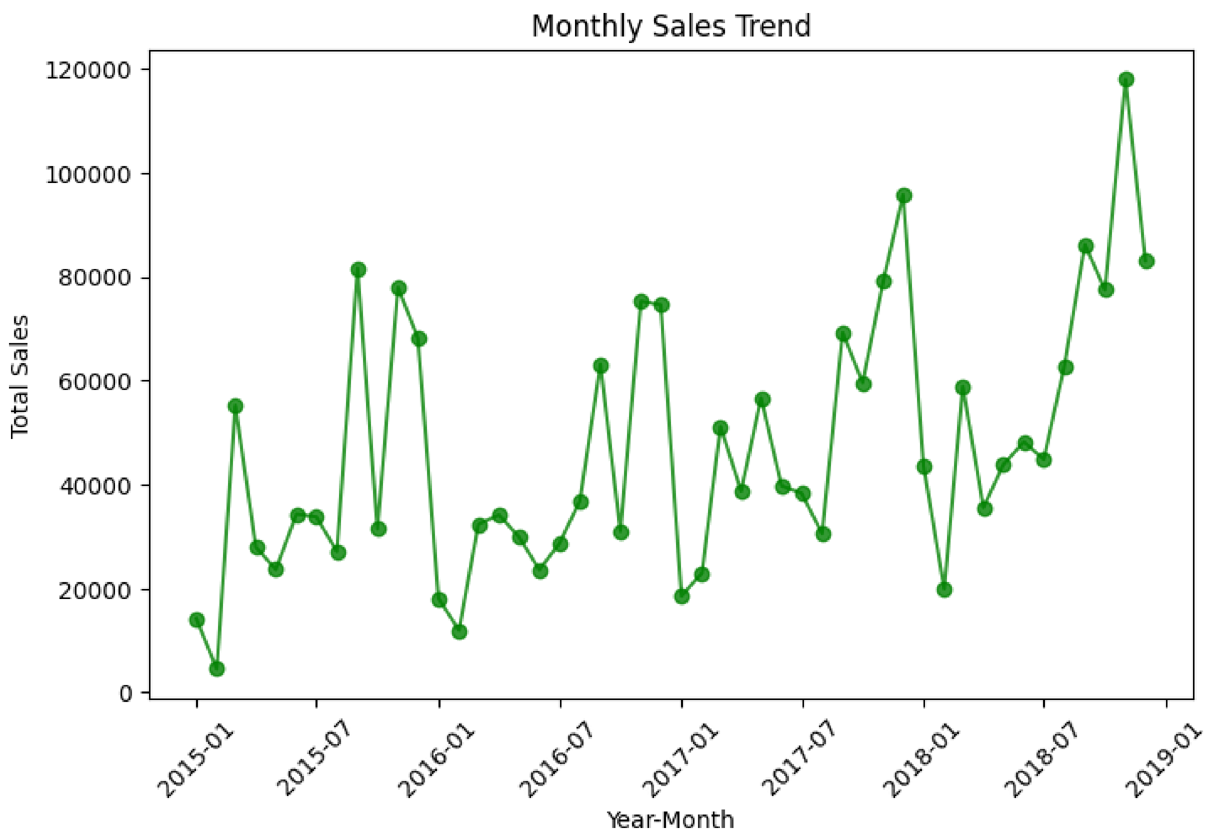
Top 10 Products Sales



04 EDA Insights

Time Series Analysis

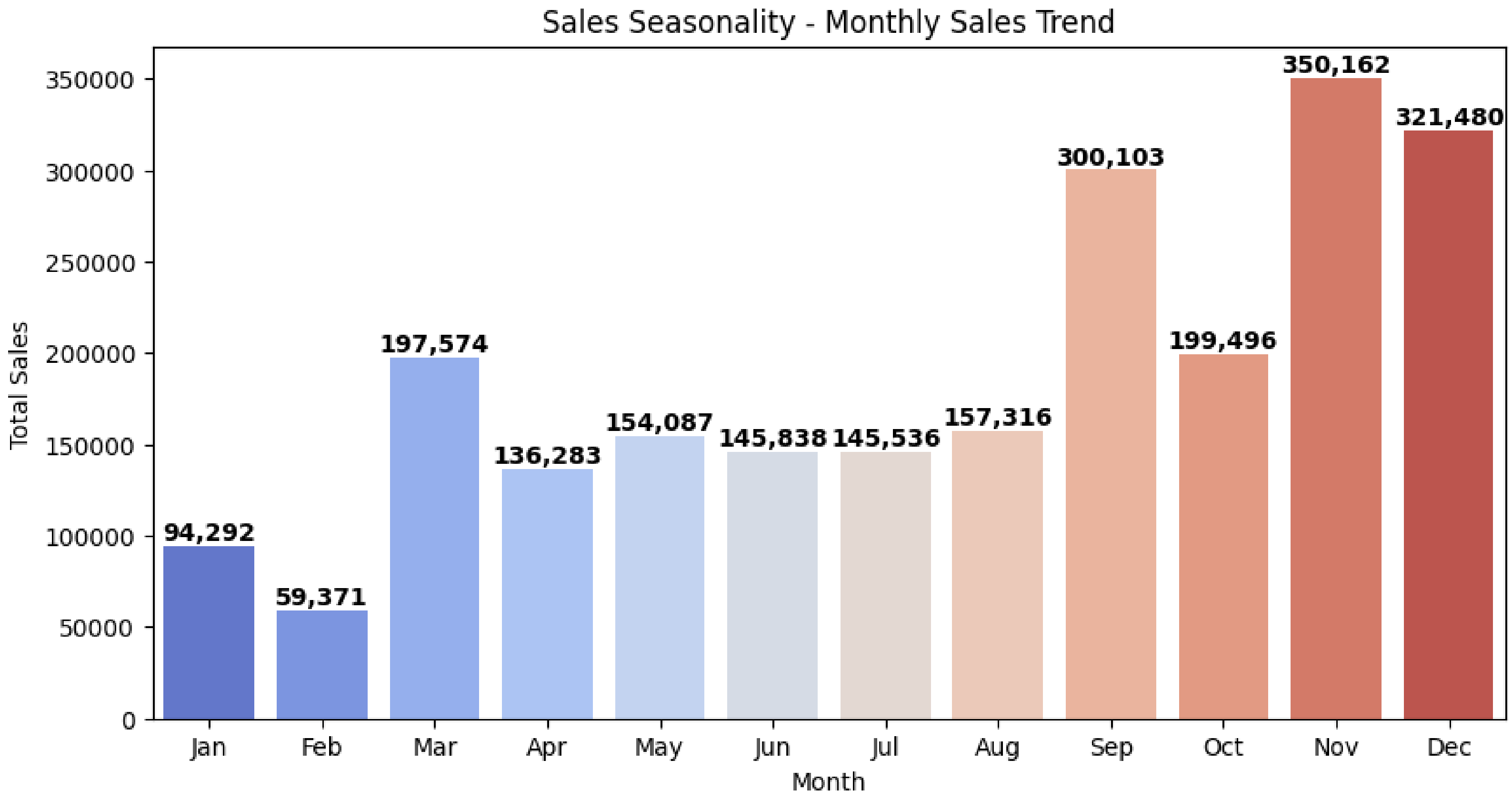
Peak sales occurred between **2018** **and** **2019.**



04 EDA Insights

Time Series Analysis

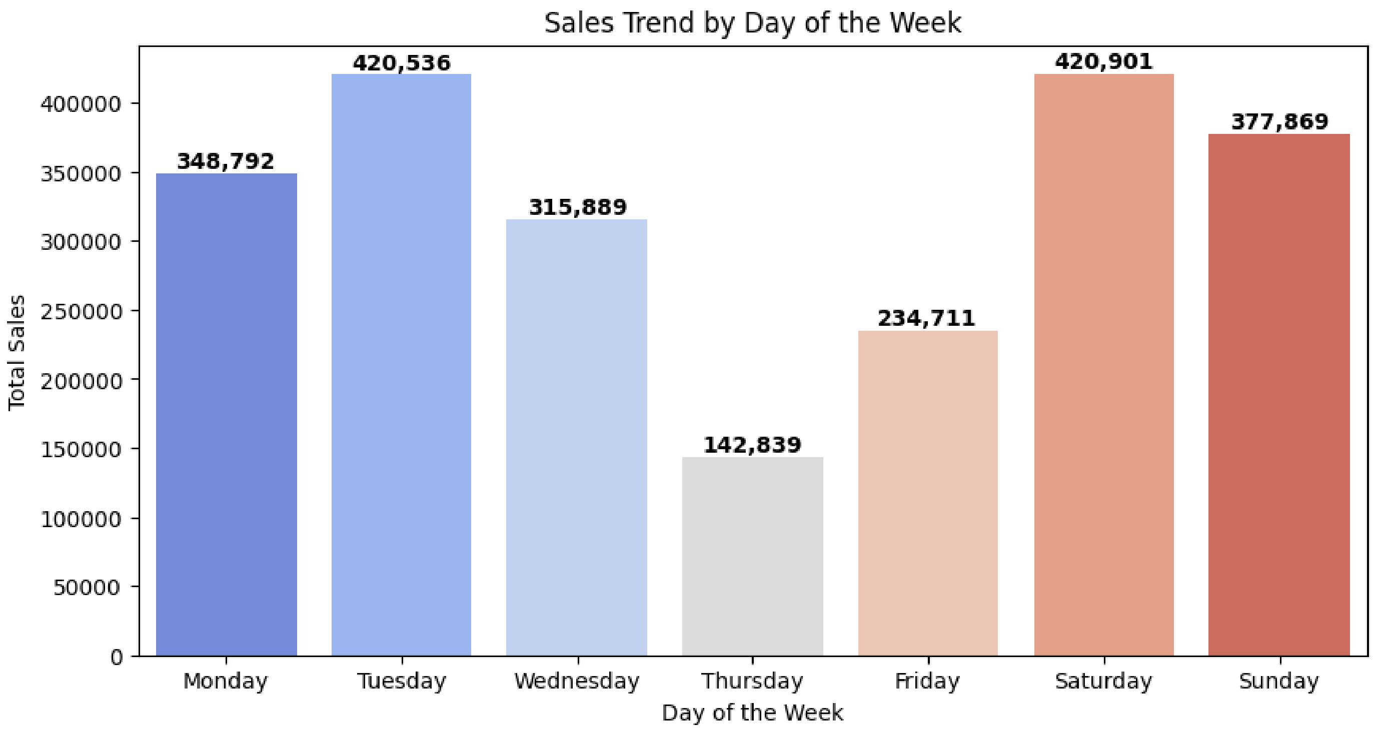
Peak sales occur in November, suggesting the need for targeted marketing and optimized inventory to maximize revenue.



04 EDA Insights

Time Series Analysis

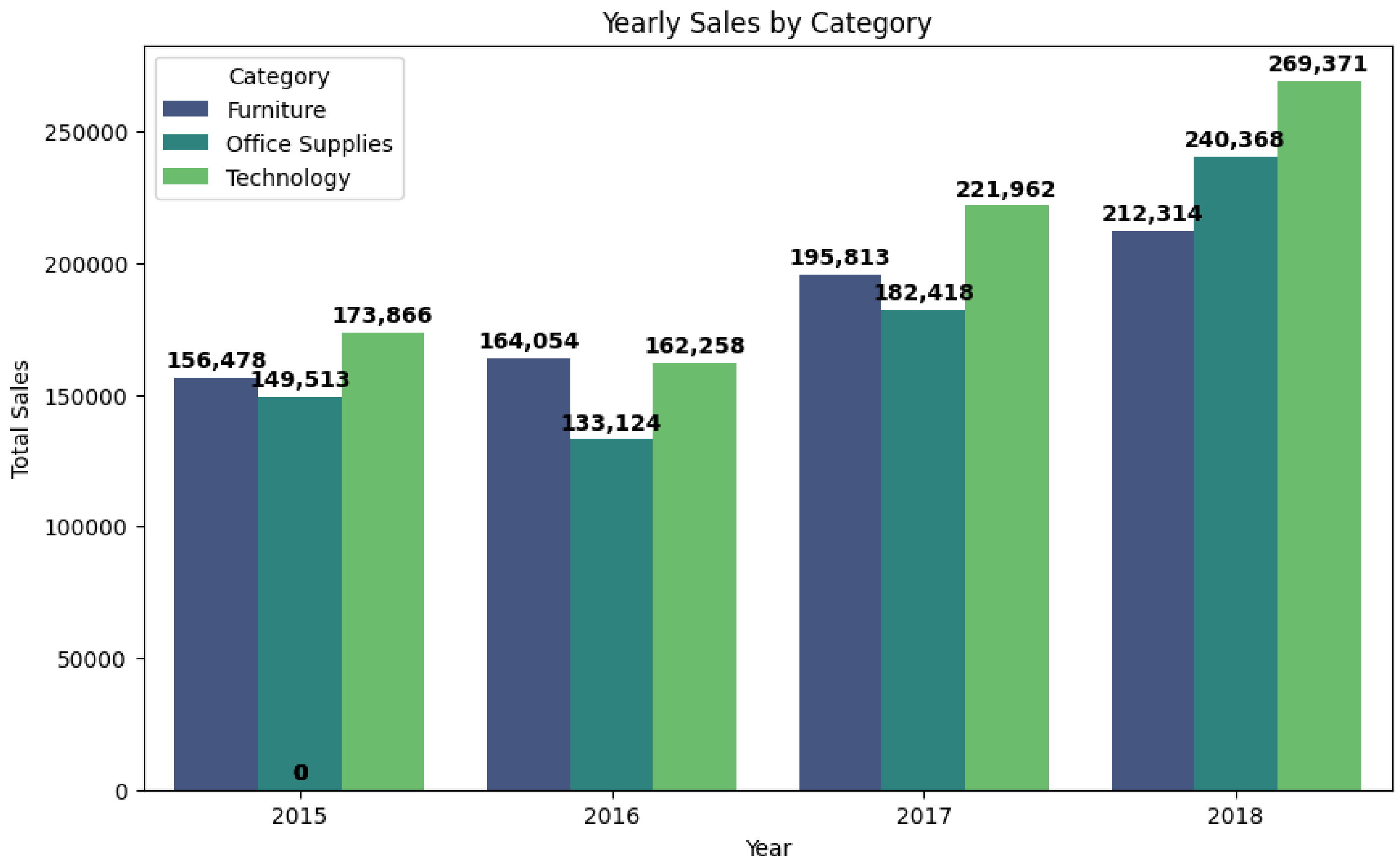
Highest sales occur on Saturdays, suggesting a strategic focus on weekend promotions and inventory management to maximize revenue.



04 EDA Insights

Time Series Analysis

Highest Sales For Tech Category, This suggests that **expanding the range of technology-related products** could lead to **higher future sales and revenue growth.**



04 EDA Insights

Geographical Analysis

Top 3 Cities in sales are

25.1% in
New York

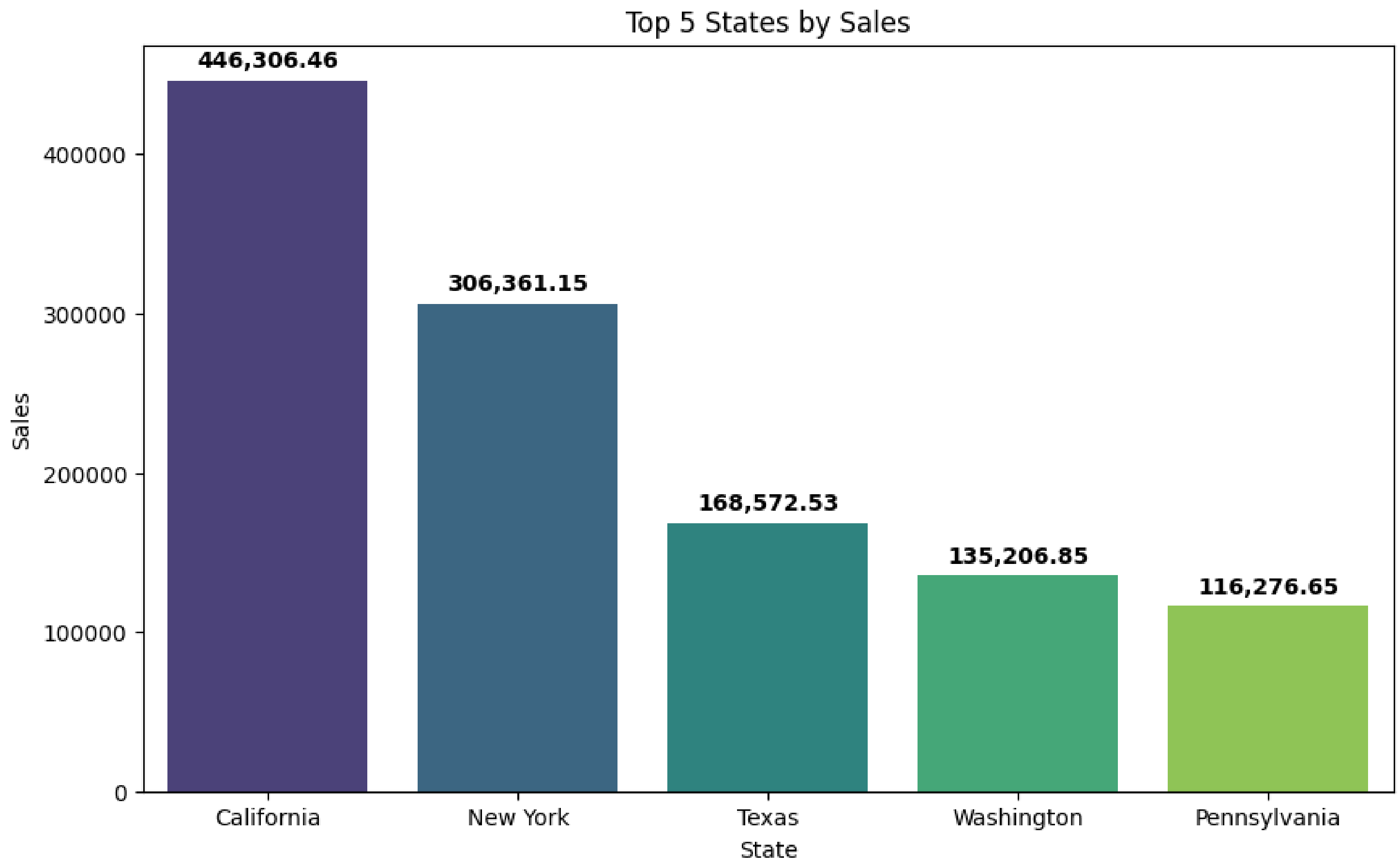
17.2% in
Los Angeles

11.5% in
Seattle

04 EDA Insights

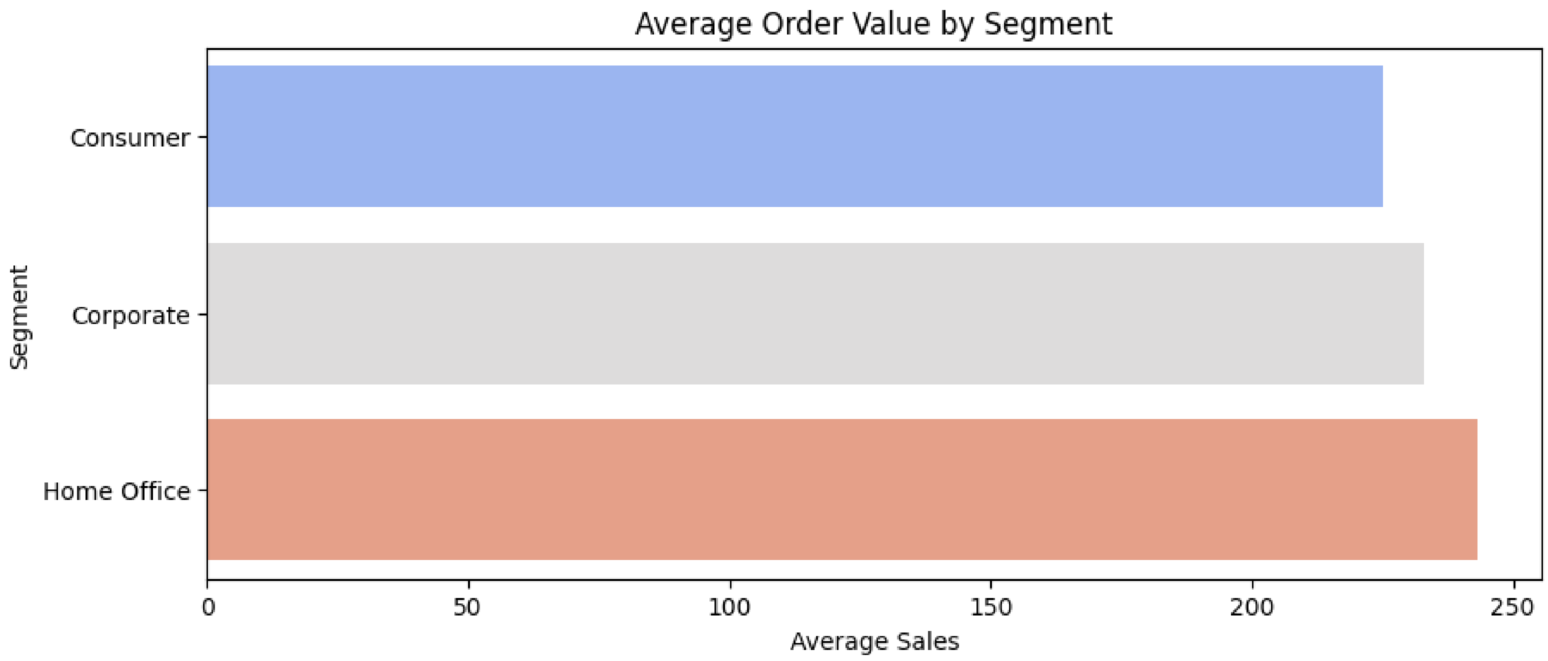
Geographical Analysis

California records the **highest sales**, indicating a strong market **presence and potential for further expansion.**



04 EDA Insights

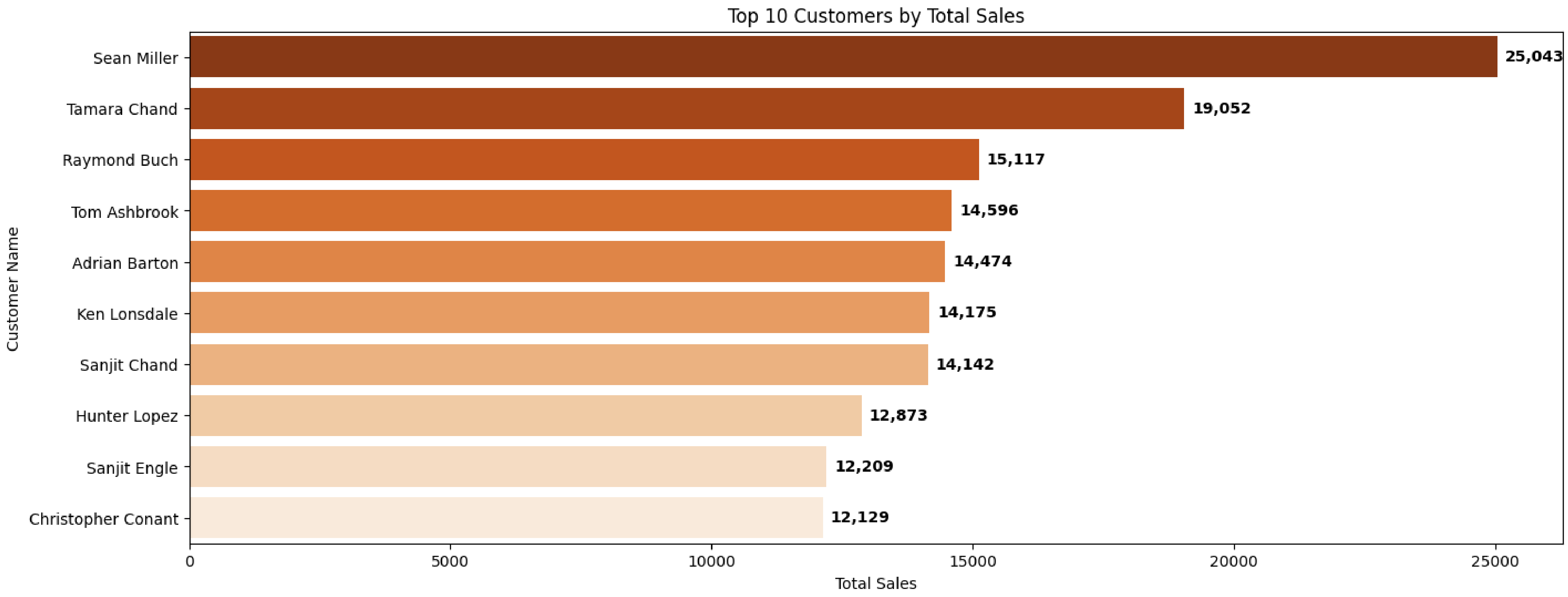
Customer Analysis



The **Home Office** segment generates **the highest sales,** suggesting **a key target audience** for future strategies

04 EDA Insights

Customer Analysis



05 Modeling

Objectives

- Select appropriate regression models
- Evaluate their performance using key metrics
- Apply cross-validation to ensure model reliability

Models Used

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

Evaluation Metrics

- MAE, MSE, RMSE: Measure prediction error
- RMSLE: Accounts for relative error, useful for skewed sales data

05 Modeling : Linear Regression

Interpretation

- The average error between predicted and actual sales is ~241 units.
- A relatively high RMSLE (1.55) suggests the model struggles more with larger sales values.

Conclusion

The linear regression model gave us a decent start, but as seen in the metrics, there's quite a bit of error — especially noticeable in the RMSLE. This suggests that our sales data might have a skewed or wide distribution, which this linear model doesn't capture well. That's why we explored more flexible models next.

Performance Metrics	
RMSLE	1.55
RMSE	740.92
MSE	548960.22
MAE	240.86

05 Modeling : Decision Tree Regressor

Interpretation

- The MAE increased to ~326 units compared to Linear Regression's 240.
- RMSLE (1.63) is higher than Linear Regression → worse at handling larger sales values.

Conclusion

The Decision Tree didn't outperform Linear Regression, indicating a need for either better tuning or an ensemble method like Random Forest.

Performance Metrics	
RMSLE	1.63
RMSE	1053.69
MSE	1110271.53
MAE	326.07

05 Modeling : Random Forest

Interpretation

- Best RMSLE of all models (1.52), suggesting it's better at predicting both low and high sales values.
- Lower RMSE and MSE than Decision Tree → better generalization.
- Slightly higher MAE than Linear Regression, but overall more balanced performance.

Conclusion

Random Forest offers the most consistent performance across all metrics. It handles complexity better than a single Decision Tree and reduces overfitting while capturing non-linear patterns.

Performance Metrics	
RMSLE	1.52
RMSE	783.14
MSE	613309.2
MAE	265.67

05 Modeling : Cross-Validation

Interpretation

- **Linear Regression** has the lowest average MAE, indicating the most consistent performance across folds.
- **Random Forest** provides a good tradeoff: lower RMSLE and RMSE than Decision Tree while keeping MAE close to Linear Regression.
- **Decision Tree** performs the worst, confirming its tendency to overfit.

Model	RMSLE	RMSE	MSE	MAE
Linear Regression	1.552	569.81	324685.17	213.69
Decision Tree	1.682	832.26	692660.93	289.28
Random Forest	1.554	668.04	4446271.71	248.64

Conclusion

Cross-validation confirms that Random Forest is the most robust overall, while Linear Regression is still surprisingly strong as a baseline.

Questions

Thank You

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